Transformer-based Kinematics-Informed Trajectory Estimator for 3D Trajectory Forecasting

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Abstract

Accurate three-dimensional trajectory forecasting for Unmanned Aerial Vehicles (UAVs) and Unmanned Underwater Vehicles (UUVs) is critical for safe autonomous navigation and collision avoidance. However, existing data-driven autoregressive models suffer from severe exposure bias and frequently generate physically implausible paths over extended horizons. To address this, we propose TransKITE (Transformer-based Kinematics-Informed Trajectory Estimator), a novel hybrid architecture that synergizes a global-context Transformer encoder with a zero-input autonomous GRU decoder to drastically mitigate step-wise error accumulation. To encourage physical realism, TransKITE augments raw positional inputs with noise-suppressed, Savitzky–Golay-filtered velocities. The model is optimized using a physics-informed objective that enforces strict discrete-time kinematic consistency through a first-order Euler local-update constraint and a macroscopic velocity-integration constraint. Extensive experiments on complex UAV and UUV datasets demonstrate that TransKITE establishes a new state-of-the-art. It achieves the lowest spatial errors across both domains and significantly extends valid prediction time over competitive baselines. Our findings prove that embedding mathematical inductive biases into sequence models profoundly enhances long-horizon stability for autonomous systems.

Keywords: 3D Trajectory Forecasting, Physics-Informed Neural Networks (PINN), Transformer-GRU Architecture, Kinematics Constraints, Autonomous Navigation

1 Introduction

Accurate 3D trajectory forecasting for Unmanned Aerial Vehicles (UAVs) and Unmanned Underwater Vehicles (UUVs) is critical to the safety of autonomous navigation systems. In real-world environments, predictors must handle noisy measurements and highly non-linear motion, challenges that are amplified in 3D settings where the additional altitude or depth dimension increases complexity.

While deep learning currently dominates trajectory forecasting, existing architectures face notable limitations. Recurrent models (LSTMs, GRUs) capture short-term patterns but suffer from exposure bias (Bengio et al. 2015), causing severe drift during extended rollouts. Conversely, Transformers (Giuliari et al. 2021) excel at capturing long-range global dependencies but lack the recursive inductive bias needed to model continuous agent kinematics. Furthermore, continuous-depth models such as Neural ODEs (Chen et al. 2018) offer elegant foundations but suffer from high computational costs and instability, hindering real-time deployment.

A major limitation of these architectures is their purely data-driven nature. By treating prediction merely as coordinate regression, they ignore physical laws, often leading to implausible outputs like sudden teleportation. Integrating physics into deep neural networks via Physics-Informed Neural Networks (PINNs) (Karniadakis et al. 2021) has shown great potential. However, effectively applying physics-guided constraints to the stochastic, noisy, and discrete-time domain of 3D UAV and UUV trajectories (Cui et al. 2020) remains an open challenge.

To bridge this gap, we propose TransKITE, a novel hybrid architecture designed for long-horizon 3D trajectory forecasting. To handle noisy sensor data, TransKITE augments spatial inputs with Savitzky–Golay-filtered velocities (Luo et al. 2005), producing a smoothed 6D kinematic state. Architecturally, TransKITE synergizes a Transformer encoder's global context-awareness with a GRU's temporal progression. The encoder distills long-term histories into a latent state bridge, initializing a zero-input autonomous GRU decoder to reduce exposure bias during autoregressive rollouts. Furthermore, a physics-informed hybrid loss enforces mathematically coherent and physically plausible trajectory evolution. Evaluated on a mixed synthetic–real 3D UAV dataset (Nacar et al. 2024) and the 3DMoTraj UUV dataset (Zhou et al. 2025), TransKITE outperforms purely data-driven recurrent, Transformer, and specialized baselines. Combining hybrid encoding, autonomous decoding, and lightweight Euler-based kinematic constraints substantially reduces positional errors and extends the valid prediction time for safe navigation. The main contributions are:

1. We propose TransKITE, a novel hybrid architecture for discrete-time 3D trajectory forecasting combining a Transformer encoder, a zero-input autonomous GRU decoder, and kinematics-informed learning. Integrating global spatiotemporal modeling with motion-aware state augmentation and Euler-based consistency constraints mitigates exposure bias while guaranteeing physically plausible predictions.
2. Extensive experiments and ablations establish TransKITE as a state-of-the-art solution achieving superior accuracy and operational reliability over extended forecasting horizons.

2 Related Work

2.1 Conventional Mathematical Models

Traditional trajectory forecasting relies heavily on explicit mathematical descriptions, including kinematic, dynamic, Kalman filtering, graph-based, and aerodynamic models (Shukla et al. 2025). These methods remain attractive due to their interpretability, computational efficiency, and seamless integration into control pipelines. In resource-constrained systems or scenarios with precise domain knowledge, they provide a principled starting point.

Among these, state-space filtering approaches are widely explored. Kalman filters and their extensions, such as IMM filters, support real-time updates and incorporate prior kinematic assumptions to track and denoise short-horizon motion (Zhu et al. 2024; Ayhan and Samet 2016; Wang et al. 2017). Another popular direction formulates forecasting as a constrained optimization problem. Here, motion is governed by explicit equations to minimize energy or avoid collisions, while graph-based formulations compute paths via constrained search strategies (Arantes et al. 2016).

The primary limitation of these models is their strict dependence on the assumed dynamics. Practical motion involves complex environmental uncertainties and nonlinear perturbations that simple closed-form equations fail to capture. When actual operating regimes deviate from theoretical assumptions, conventional methods degrade rapidly, motivating the shift toward machine learning.

2.2 Data-Driven Machine Learning Models

To overcome the rigid assumptions of mathematical models, contemporary research has increasingly adopted deep learning to extract motion patterns directly from historical data. Recurrent neural networks, such as LSTMs, quickly became strong baselines for extracting sequential patterns without explicit equations (Zhang et al. 2022; Yang et al. 2023; Silvestre et al. 2024). However, as noted earlier, standard autoregressive recurrent architectures suffer from exposure bias (Bengio et al. 2015), leading to severe trajectory drift over long horizons.

To capture broader temporal contexts, higher-capacity architectures have emerged (Jia et al. 2022). Transformer-based models leverage self-attention for global temporal modeling, while works like SCA-LSTM (Zhou et al. 2025) explicitly model 3D cross-axis dependencies. Concurrently, Neural ODEs (Chen et al. 2018) offer an elegant continuous-time formulation by directly modeling the hidden state's derivative. Nevertheless, purely data-driven Transformers struggle to encode physical inductive biases, while Neural ODEs face high computational overhead. Ultimately, purely data-driven networks treat prediction merely as statistical coordinate regression; by disregarding kinematic laws, they often produce outputs that violate basic physical relationships, such as spatial displacements contradicting velocity vectors.

2.3 Physics-Informed and Kinematics-Aware Models

To ensure long-horizon plausibility, recent literature integrates physical knowledge into deep learning. Physics-Informed Neural Networks (PINNs) (Karniadakis et al. 2021)

directly encode governing dynamics into the training objective. For instance, PINNs have been employed to learn quadrotor dynamics by embedding known equations of motion, yielding dynamically feasible paths (Gu et al. 2024; Serrano et al. 2024). While providing strong guarantees, these methods require knowledge of the underlying differential equations, limiting their applicability to generalized, noisy 3D trajectory data where exact forces are unknown. Another direction is kinematics-informed modeling, which incorporates basic first-order motion structures. The VECTOR architecture (Nacar et al. 2024), for example, uses historical velocity estimates as primary inputs rather than relying solely on position. However, merely feeding velocity as an input signal does not mathematically constrain the network's future coordinate generations.

Methodology	**Spatiotemporal Global Attention**	**Exposure-Bias Free Generation**	**Explicit Kinematic Constraints**	**Denoised Kinematic Input**
Standard LSTM/GRU	No	No	No	No
PatchTST / Informer	Yes	No	No	No
PINN-based (Gu et al. 2024)	No	No	Yes	No
VECTOR (Nacar et al. 2024)	No	No	No	Yes (Raw diff.)
TransKITE (Ours)	**Yes**	**Yes (Zero-Input)**	**Yes (Euler-based)**	**Yes (Savitzky-Golay)**

Table 1: Architectural comparison of TransKITE with existing frameworks.

As summarized in Table 1, TransKITE advances this paradigm by bridging PINN-inspired guidance, global attention, and recurrent modeling. Unlike previous methods relying on noisy finite differences, TransKITE robustly extracts kinematics via a Savitzky–Golay filter. It utilizes a Transformer encoder for spatiotemporal context but delegates generation to a *zero-input autonomous GRU decoder*, neutralizing exposure bias. Most importantly, TransKITE enforces explicit mathematical realism: by binding predicted spatial and velocity sequences using a first-order Euler local-update constraint and a global integration consistency constraint, it seamlessly embeds the strictness of PINNs into a lightweight, highly generalizable forecasting framework.

3 Methods

We consider forecasting trajectories in three-dimensional space, where each position is represented by longitude, latitude, and a vertical coordinate. At each discrete time step t, with a sampling interval Δt, the position is represented as $\mathbf{p}_t \in \mathbb{R}^3$. Given a sequence of T historical positions $\{\mathbf{p}_k\}_{k=t-T+1}^{t}$, the objective is to predict the next F future positions $\{\hat{\mathbf{p}}_{t+k}\}_{k=1}^{F}$.

During training, trajectory data are segmented into windows of length $T+F$. Without overlap, a trajectory of length N yields $\left\lfloor \frac{N}{T+F} \right\rfloor$ training samples. By introducing a stride $S \ll (T+F)$, we construct overlapping windows, increasing the number of samples to $\left\lfloor \frac{N-(T+F)}{S} \right\rfloor + 1$. This sliding-window strategy improves data efficiency by exposing the model to multiple shifted temporal contexts. During inference, each trajectory is initialized using its first T observed positions. Based on this historical

context, the model predicts the next F future positions. These predictions are then appended to the sequence, and the context window of length T is shifted forward by F steps. This iterative autoregressive rollout allows the model to produce long-horizon trajectory forecasts seamlessly beyond a single prediction horizon. To address the compounding drift typical of extended autoregressive rollouts, we propose TransKITE. The macroscopic workflow of the architecture is illustrated in Figure 1.

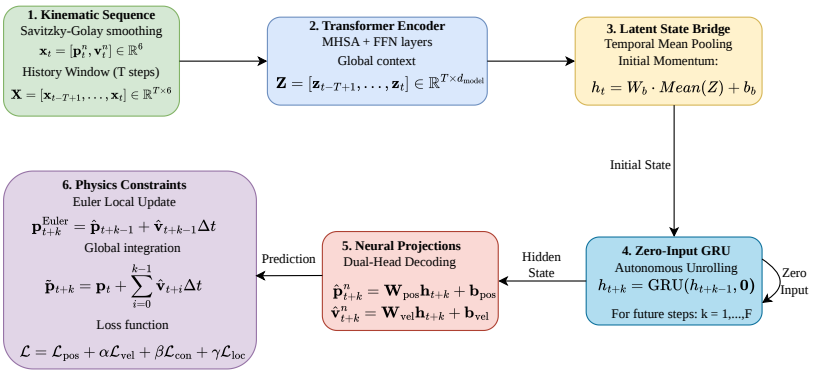


Fig. 1: Overall macroscopic workflow of TransKITE.

The pipeline operates in six main stages: first, the raw trajectory is augmented into a smoothed kinematic sequence (Block 1) and processed by a Transformer encoder to extract global spatiotemporal context (Block 2). A latent state bridge compresses this history into a robust initial momentum vector (Block 3). For the forecasting phase, a zero-input autonomous GRU mitigates exposure bias by unrolling future states without relying on noisy intermediate coordinate inputs (Block 4). These hidden states are decoded into spatial and kinematic outputs via dual neural projections (Block 5), which are finally bounded by strict physics-informed kinematic constraints during optimization (Block 6). Detailed theoretical analyses and mathematical proofs regarding the error bounds of this zero-input decoder and global kinematic consistency are provided in Section 4.

3.1 Problem Formulation and Velocity-Aware Augmentation

To enrich the model input with explicit motion dynamics, we augment the raw position sequence with estimated velocities. Instead of using naive finite differences - which dramatically amplify the high-frequency measurement noise inherent in pose observations - we apply a Savitzky–Golay differentiation filter. This filter suppresses noise while preserving the underlying continuous motion trend and kinematic smoothness.

Specifically, for each time step, we consider a short sliding window of size $L = 11$ - a value chosen empirically to balance high-frequency noise suppression with the preservation of rapid kinematic changes (see Appendix A for a detailed sensitivity analysis) - fit a local 3rd-order polynomial to the positions via linear least squares,

and compute its analytical derivative at the center. Let $L = 2m + 1$ (where $m = 5$). This procedure is mathematically equivalent to applying a discrete convolution over the position sequence with pre-computed coefficients d_j. To correctly map this to the physical time domain, the convolution is scaled by the sampling interval Δt:

$$\mathbf{v}_t = \frac{1}{\Delta t} \sum_{j=-m}^{m} d_j \mathbf{p}_{t+j}. \tag{1}$$

For the causal boundaries of the sequence during online inference (where future positions $t + j$ are unobserved), the filter dynamically switches to one-sided polynomial extrapolation, ensuring that real-time causality is strictly maintained.

Both positions and estimated velocities are subsequently normalized using their respective maximum ℓ^2-norms computed exclusively over the training set $\mathcal{D}_{\text{train}}$. This ensures consistent scaling and stable gradient updates:

$$\mathbf{p}_t^n = \frac{\mathbf{p}_t}{\max\limits_{\mathbf{p} \in \mathcal{D}_{\text{train}}} \|\mathbf{p}\|_2}, \quad \mathbf{v}_t^n = \frac{\mathbf{v}_t}{\max\limits_{\mathbf{v} \in \mathcal{D}_{\text{train}}} \|\mathbf{v}\|_2}. \tag{2}$$

The final augmented kinematic state input to the TransKITE model at each time step is defined as the concatenation of the normalized position and velocity:

$$\mathbf{x}_t = [\mathbf{p}_t^n, \mathbf{v}_t^n] \in \mathbb{R}^6. \tag{3}$$

3.2 TransKITE: Hybrid Transformer-GRU Architecture

To model the complex spatiotemporal dependencies of 3D trajectories, we propose TransKITE, a hybrid architecture that synergizes the global context-awareness of Transformers with the sequential inductive bias of Gated Recurrent Units (GRUs). The model consists of a Transformer-based encoder, a latent state bridge, and a zero-input GRU decoder equipped with dual output heads.

3.2.1 Transformer Encoder

The encoder takes the sequence of augmented kinematic states $\mathbf{X} = [\mathbf{x}_{t-T+1}, \ldots, \mathbf{x}_t] \in \mathbb{R}^{T \times 6}$ as input. To capture long-term global dependencies without the sequential bottleneck of standard recurrent networks, we employ a Multi-Head Self-Attention (MHSA) mechanism. First, the input sequence is linearly projected into a high-dimensional feature space $\mathbb{R}^{d_{\text{model}}}$, and positional encodings are added to retain temporal order information. The sequence is processed by N_e stacked Transformer encoder layers. Each layer comprises an MHSA module followed by a position-wise Feed-Forward Network (FFN), with residual connections and layer normalization applied around each sub-layer. The output is a highly contextualized feature sequence $\mathbf{Z} = [\mathbf{z}_{t-T+1}, \ldots, \mathbf{z}_t] \in \mathbb{R}^{T \times d_{\text{model}}}$, encapsulating comprehensive flight history, aerodynamic trends, and complex cross-timestep interactions.

3.2.2 Latent State Bridge

Traditional sequence-to-sequence models often struggle with exposure bias during long-horizon autoregressive forecasting. To mitigate this, TransKITE transitions from a token-by-token sequence representation to an autonomous dynamical system formulation. We introduce a latent state bridge that compresses the entire spatiotemporal memory $\mathbf{Z}$ into a dense initial hidden state $\mathbf{h}_t$.

Crucially, rather than flattening the sequence, which destroys temporal invariance, we apply temporal mean pooling over the encoder outputs, followed by a linear projection to map the context into the multi-layer GRU's hidden space:

$$\mathbf{h}_t = \text{Reshape}\left(\mathbf{W}_{\text{bridge}}\left[\frac{1}{T}\sum_{i=t-T+1}^{t}\mathbf{z}_i\right] + \mathbf{b}_{\text{bridge}}\right). \tag{4}$$

This compact vector $\mathbf{h}_t$ acts as the initial "momentum" for the forecasting phase, effectively distilling the necessary kinematic information into a robust, unified representation.

3.2.3 GRU Decoder and Dual Output Heads

For the decoding phase, we utilize a GRU (Cho et al. 2014). However, instead of using standard teacher forcing or continuously feeding previous predictions back as inputs - which inevitably leads to severe trajectory drift - we design the decoder as a *zero-input autonomous dynamical system*.

The decoder GRU is solely initialized with the latent state $\mathbf{h}_t$ provided by the bridge. At each future prediction step $k \in \{1, \ldots, F\}$, the GRU receives a zero-vector input $\mathbf{0}$ and updates its hidden state purely based on internal transitions. Consequently, the standard GRU gating mechanism is mathematically simplified to:

$$\mathbf{r}_{t+k} = \sigma(\mathbf{W}_{hr}\mathbf{h}_{t+k-1} + \mathbf{b}_{hr}), \tag{5}$$
$$\mathbf{z}_{t+k} = \sigma(\mathbf{W}_{hz}\mathbf{h}_{t+k-1} + \mathbf{b}_{hz}), \tag{6}$$
$$\mathbf{n}_{t+k} = \tanh(\mathbf{r}_{t+k} \odot (\mathbf{W}_{hn}\mathbf{h}_{t+k-1} + \mathbf{b}_{hn})), \tag{7}$$
$$\mathbf{h}_{t+k} = (1 - \mathbf{z}_{t+k}) \odot \mathbf{n}_{t+k} + \mathbf{z}_{t+k} \odot \mathbf{h}_{t+k-1}, \tag{8}$$

where $\sigma(\cdot)$ denotes the sigmoid function, and $\odot$ represents the Hadamard product. By isolating the decoder from raw coordinate inputs at each step, the network is forced to unroll the trajectory forward by exclusively relying on the robust dynamics encoded in its hidden state, thereby drastically extending its Valid Time (T_v) over long predictive horizons. Finally, to enable simultaneous prediction of position and velocity, we employ a dual-head output design. Each generated hidden state $\mathbf{h}_{t+k}$ is passed through two separate linear projections:

$$\hat{\mathbf{p}}^n_{t+k} = \mathbf{W}_{\text{pos}}\mathbf{h}_{t+k} + \mathbf{b}_{\text{pos}}, \quad \hat{\mathbf{v}}^n_{t+k} = \mathbf{W}_{\text{vel}}\mathbf{h}_{t+k} + \mathbf{b}_{\text{vel}}. \tag{9}$$

This design allows the shared recurrent backbone to maintain a unified temporal dynamic, while the specialized heads accurately decode the specific spatial positions and kinematic velocities without mutual interference.

3.3 Physics-Informed Kinematic Constraints

While the hybrid Transformer-GRU architecture effectively learns spatiotemporal patterns, purely data-driven predictions often produce physically inconsistent outputs during long-horizon autoregressive rollouts. Independently regressed positions and velocities frequently violate basic kinematic relationships. To address this, we incorporate physics-informed constraints that explicitly enforce strict dynamic consistency on the network's outputs. The fundamental supervised learning objective minimizes the discrepancy between the network's normalized predictions and the ground truth. We define the data-driven loss for position and velocity as:

$$\mathcal{L}_{\text{pos}} = \text{MSE}(\hat{\mathbf{p}}^n_{t+k}, \mathbf{p}^n_{t+k}), \quad \mathcal{L}_{\text{vel}} = \text{MSE}(\hat{\mathbf{v}}^n_{t+k}, \mathbf{v}^n_{t+k}). \tag{10}$$

To regularize the model beyond matching raw data points, we introduce two physics-guided constraints formulated in the denormalized, real-world physical space (i.e., meters and meters per second).

First, a local kinematic constraint enforces step-wise consistency by relating consecutive predictions. Mathematically, this constraint is derived from the first-order Taylor series expansion of the continuous position function $\mathbf{p}(t+\Delta t) = \mathbf{p}(t)+\mathbf{p}'(t)\Delta t+\mathcal{O}(\Delta t^2)$. By truncating the higher-order acceleration terms under the assumption of a sufficiently small Δt, we adopt the first-order Euler method to balance numerical stability with the discrete nature of neural network outputs. Given the inherently noisy dynamics of real-world motion, the Euler method provides a computationally efficient and empirically robust constraint without introducing the complex optimization landscapes or interpolation artifacts associated with higher-order integrators. Thus, the physically updated position is computed as: $\mathbf{p}^{\text{Euler}}_{t+k} = \hat{\mathbf{p}}_{t+k-1} + \hat{\mathbf{v}}_{t+k-1}\Delta t$.

Second, a velocity-integration consistency constraint reconstructs an alternative position trajectory using solely the predicted velocities, accumulating from the last observed ground-truth position: $\tilde{\mathbf{p}}_{t+k} = \mathbf{p}_t + \sum_{i=0}^{k-1} \hat{\mathbf{v}}_{t+i}\Delta t$. This macroscopic term prevents the sequence of predicted velocities from drifting away from the global spatial displacement over time.

A critical challenge in Physics-Informed Neural Networks (PINNs) is gradient imbalance, where physical losses calculated in large real-world scales (e.g., thousands of meters) overwhelm the normalized data losses (bounded between 0 and 1). To ensure stable optimization, we re-normalize the physically derived positions before computing their Mean Squared Error against the normalized network predictions:

$$\mathcal{L}_{\text{loc}} = \text{MSE}\left(\hat{\mathbf{p}}^n_{t+k}, \frac{\mathbf{p}^{\text{Euler}}_{t+k}}{\max\limits_{\mathbf{p}\in\mathcal{D}_{train}} \|\mathbf{p}\|_2}\right), \quad \mathcal{L}_{\text{con}} = \text{MSE}\left(\hat{\mathbf{p}}^n_{t+k}, \frac{\tilde{\mathbf{p}}_{t+k}}{\max\limits_{\mathbf{p}\in\mathcal{D}_{train}} \|\mathbf{p}\|_2}\right). \tag{11}$$

The final training objective synergizes the data-driven and physics-informed components:

$$\mathcal{L} = \mathcal{L}_{\text{pos}} + \alpha\mathcal{L}_{\text{vel}} + \beta\mathcal{L}_{\text{con}} + \gamma\mathcal{L}_{\text{loc}}, \tag{12}$$

where α, β, and γ are hyperparameters carefully tuned to maintain a balance between learning from observed data patterns and adhering to fundamental kinematic laws.

4 Theoretical Analysis

4.1 Error Accumulation in TransKITE

In standard autoregressive sequence-to-sequence models, the decoder takes its own previous prediction $\hat{\mathbf{x}}_{t+k-1}$ as input for the next step. This introduces a compounding error phenomenon known as exposure bias. In this section, we theoretically demonstrate how the zero-input autonomous GRU decoder in TransKITE mathematically bounds this error propagation.

Theorem 1 *(Error Bound of Zero-Input Decoder). Let the hidden state transition function be uniformly Lipschitz continuous with constant $L_h \in (0, 1)$. Given an initial state error δ_0 and a bounded step-wise input noise ϵ_{in}, the hidden state error of a standard autoregressive decoder accumulates as $\mathcal{O}\big(L_h^k\delta_0 + \epsilon_{in}\sum_{i=0}^{k-1} L_h^i\big)$. In contrast, within a single prediction block, the zero-input GRU completely eliminates intra-block spatial noise injection ($\epsilon_{in} = 0$), reducing the local hidden state error bound to a strict exponential decay: $\delta_k \leq L_h^k\delta_0$.*

Proof Let the standard hidden state update function be defined as $\mathbf{h}_{t+k} = f(\mathbf{h}_{t+k-1}, \mathbf{x}_{t+k})$. Assume f is uniformly Lipschitz continuous with respect to both the hidden state and the input, with Lipschitz constants $L_h \in (0, 1)$ and $L_x > 0$, respectively.

For a standard autoregressive decoder, the predicted hidden state relies on the previously predicted (and therefore noisy) input $\hat{\mathbf{x}}_{t+k}$. The error at step k, denoted as $\delta_k = \|\hat{\mathbf{h}}_{t+k} - \mathbf{h}_{t+k}\|$, can be bounded using the triangle inequality and the Lipschitz definition:

$$\delta_k = \|f(\hat{\mathbf{h}}_{t+k-1}, \hat{\mathbf{x}}_{t+k}) - f(\mathbf{h}_{t+k-1}, \mathbf{x}_{t+k})\| \leq L_h\|\hat{\mathbf{h}}_{t+k-1} - \mathbf{h}_{t+k-1}\| + L_x\|\hat{\mathbf{x}}_{t+k} - \mathbf{x}_{t+k}\|.$$

Let $\|\hat{\mathbf{x}}_{t+k} - \mathbf{x}_{t+k}\| \leq \epsilon_{in}$ be the upper bound of the input error at any given step. Equation 13 establishes a recursive relationship: $\delta_k \leq L_h\delta_{k-1} + L_x\epsilon_{in}$.

Unrolling this recursion from step k down to the initial step 0 yields:

$$\delta_k \leq L_h^k\delta_0 + L_x\epsilon_{in}\sum_{i=0}^{k-1} L_h^i. \tag{13}$$

As $k \to \infty$, even with a stable hidden dynamic ($L_h < 1$), the geometric series converges to a non-zero persistent error $\frac{L_x\epsilon_{in}}{1-L_h}$. This mathematically proves the vulnerability of standard models to compounding exposure bias.

For the TransKITE zero-input decoder, the input is strictly locked to the zero-vector $\mathbf{0}$ at all future steps. Thus, the actual and predicted transitions simplify to $\mathbf{h}_{t+k} = f(\mathbf{h}_{t+k-1}, \mathbf{0})$ and $\hat{\mathbf{h}}_{t+k} = f(\hat{\mathbf{h}}_{t+k-1}, \mathbf{0})$. The input error term is perfectly nullified ($\|\mathbf{0} - \mathbf{0}\| = 0$). The recursive bound from Eq. 13 collapses to: $\delta_k \leq L_h\delta_{k-1}$. Unrolling this recursion directly yields $\delta_k \leq L_h^k\delta_0$. Since $L_h < 1$, the limit of the error $\lim_{k\to\infty}\delta_k = 0$. □

Remark 1 By architecturally enforcing a zero-input transition, TransKITE insulates the temporal unrolling from step-wise spatial noise during the inner-loop generation phase. While this guarantees the suppression of intra-block exposure bias, we acknowledge that outer-loop sliding window updates will eventually re-introduce historical noise. However, bounding the inner-loop drift inherently shifts the predictive burden entirely to the representation capacity of the initial state. This mathematically justifies our architectural design choice: employing a highly expressive Transformer encoder to synthesize a comprehensive, information-dense initial momentum $\mathbf{h}_t$ capable of sustaining long-horizon autonomous generation.

4.2 Kinematic Consistency Guarantee of the Hybrid Objective

We theoretically demonstrate that the local kinematic constraint ($\mathcal{L}_{\text{loc}}$) acts as a strict mathematical regularizer, bounding global spatial drift between dual-head predictions.

Theorem 2 *(Global Drift Bound of Local Constraints). Assume the local-update constraint $\mathcal{L}_{loc}$ bounds the step-wise kinematic error between the direct prediction and the Euler-updated position by $\epsilon > 0$. Then, the macroscopic spatial divergence between the directly predicted endpoint $\hat{\mathbf{p}}_{t+F}$ and the velocity-integrated endpoint $\tilde{\mathbf{p}}_{t+F}$ is linearly bounded by $F \cdot \epsilon$.*

Proof Let the autoregressive rollout initiate from the last observed ground-truth position, i.e., $\hat{\mathbf{p}}_t \equiv \mathbf{p}_t$. By the theorem's premise, for any future step $i \in \{1, \dots, F\}$, the deviation from the first-order Euler update is strictly bounded:

$$\|\hat{\mathbf{p}}_{t+i} - (\hat{\mathbf{p}}_{t+i-1} + \hat{\mathbf{v}}_{t+i-1}\Delta t)\|_2 \leq \epsilon. \tag{14}$$

The globally integrated position at the final horizon F is defined by accumulating predicted velocities starting from $\mathbf{p}_t$:

$$\tilde{\mathbf{p}}_{t+F} = \mathbf{p}_t + \sum_{i=0}^{F-1} \hat{\mathbf{v}}_{t+i}\Delta t. \tag{15}$$

To evaluate the macroscopic divergence $\|\hat{\mathbf{p}}_{t+F} - \tilde{\mathbf{p}}_{t+F}\|_2$, we express the directly predicted endpoint $\hat{\mathbf{p}}_{t+F}$ as a telescoping sum originating from $\mathbf{p}_t$:

$$\hat{\mathbf{p}}_{t+F} = \mathbf{p}_t + \sum_{i=1}^{F} (\hat{\mathbf{p}}_{t+i} - \hat{\mathbf{p}}_{t+i-1}). \tag{16}$$

Aligning the summation indices (i from 1 to F), the discrepancy vector simplifies into a single combined summation:

$$\hat{\mathbf{p}}_{t+F} - \tilde{\mathbf{p}}_{t+F} = \sum_{i=1}^{F} (\hat{\mathbf{p}}_{t+i} - \hat{\mathbf{p}}_{t+i-1}) - \sum_{i=1}^{F} \hat{\mathbf{v}}_{t+i-1}\Delta t = \sum_{i=1}^{F} \left[\hat{\mathbf{p}}_{t+i} - (\hat{\mathbf{p}}_{t+i-1} + \hat{\mathbf{v}}_{t+i-1}\Delta t)\right].$$

Applying the generalized triangle inequality yields:

$$\|\hat{\mathbf{p}}_{t+F} - \tilde{\mathbf{p}}_{t+F}\|_2 \leq \sum_{i=1}^{F} \left\|\hat{\mathbf{p}}_{t+i} - (\hat{\mathbf{p}}_{t+i-1} + \hat{\mathbf{v}}_{t+i-1}\Delta t)\right\|_2. \tag{17}$$

Substituting the bound from Equation 14 provides the final result:

$$\|\hat{\mathbf{p}}_{t+F} - \tilde{\mathbf{p}}_{t+F}\|_2 \leq \sum_{i=1}^{F} \epsilon = F \cdot \epsilon. \tag{18}$$

□

Remark 2 This rigorously proves that regulating the step-wise local-update constraint ($\mathcal{L}_{\text{loc}}$) inherently caps macroscopic trajectory divergence. It mathematically guarantees that the network's spatial predictions remain safely synchronized with its generated kinematic velocity vectors over extended horizons.

5 Experiment

5.1 Dataset

We utilize two datasets to evaluate the proposed model across UAV and UUV forecasting tasks. For UAV forecasting, we use the dataset introduced by Nacar et al. (2024), comprising both synthetic and real-world 3D flights. The synthetic subset is generated in Gazebo via PX4 Autopilot, following predefined patterns (e.g., circular, figure-eight), while the real-world data is aggregated from the public UZH-FPV and Mid-Air datasets. This composite dataset ensures diverse flight behaviors under varying aerodynamic conditions, enhancing the generalizability of trajectory predictors (Nacar et al. 2024). Trajectories are split into training, validation, and test sets using an 80/10/10 ratio, stratified to ensure each subset contains a representative mixture of simulated and real-world dynamics. To construct supervised samples, we apply a sliding window strategy with a historical observation length of $T = 50$, a prediction horizon of $F = 20$, and a stride of $S = 10$.

For UUV forecasting, we use the 3DMoTraj dataset (Zhou et al. 2025), which comprises 8 distinct underwater scenes. The dataset captures continuous 3D spatial transitions in highly non-linear aquatic environments, providing a challenging benchmark for modeling underwater motion. For this dataset, supervised samples are generated using a sliding window strategy with an observation window of length $T = 8$, a prediction horizon of $F = 12$, and a stride of $S = 2$.

5.2 Experimental Setup

TransKITE employs a hybrid Transformer-GRU architecture. Based on a validation grid search, the hidden dimension for both the Transformer encoder and the autonomous GRU decoder is set to 64. The model is optimized using AdamW with an initial learning rate of 10^{-3} and weight decay for regularization. A step learning rate scheduler decays the learning rate by a factor of 0.5 every 50 epochs to ensure stable convergence. The physics-guided loss penalty weights (α, β, γ) are empirically tuned to seamlessly align the kinematic constraints with the data-driven spatial loss. The model is trained for up to 500 epochs, incorporating an early stopping patience of 25 epochs monitored via the validation RMSE. Due to the distinct fluid dynamic properties of air (aerodynamics) and water (hydrodynamics), as well as different sampling frequencies, hyperparameter configurations - including sequence lengths and loss penalty weights - were optimized independently for the UAV and UUV domains.

To rigorously evaluate TransKITE, we benchmark it against a highly competitive set of baseline architectures spanning multiple paradigms: standard sequence models (LSTM), linear forecasters (DLinear), modern long-range Transformers (Informer,

PatchTST), continuous-time systems (NeuralODE), and state-of-the-art trajectory-specialized networks (TrajCLIP, MS-TIP, MRGTraj, S-Implicit, and SCA-LSTM). This provides a comprehensive evaluation across both general time-series forecasting and specialized spatial prediction frameworks.

5.3 Evaluation Setup

To comprehensively assess predictive accuracy and operational reliability, we employ multiple complementary metrics. Let $\mathbf{P} = \{\mathbf{p}_{t+k}\}_{k=1}^{F}$ and $\hat{\mathbf{P}} = \{\hat{\mathbf{p}}_{t+k}\}_{k=1}^{F}$ denote the ground-truth and predicted trajectories over the future horizon F. The instantaneous positional error at step k is defined by the ℓ^2-norm: $e_k = \|\mathbf{p}_{t+k} - \hat{\mathbf{p}}_{t+k}\|_2$.

We report three standard metrics to evaluate prediction quality:

- **Average Displacement Error (ADE):** $\text{ADE} = \frac{1}{F}\sum_{k=1}^{F} e_k$
- **Root Mean Square Error (RMSE):** $\text{RMSE} = \sqrt{\frac{1}{F}\sum_{k=1}^{F} e_k^2}$
- **Final Displacement Error (FDE):** $\text{FDE} = e_F$

Beyond error magnitude, we define **Valid Time** (T_v) to measure operational reliability during an extended continuous rollout. Let K_{max} denote the maximum evaluation horizon (e.g., $K_{max} = 120$ steps for our extended experiments). T_v represents the maximum number of consecutive future steps where the prediction remains within a strict safety threshold δ (set to 2.0 m). If $e_1 \geq \delta$, the prediction is immediately considered unreliable ($T_v = 0$). Otherwise:

$$T_v = \max\{k \in \{1, \ldots, K_{max}\} \mid e_i < \delta, \forall i \leq k\}. \tag{19}$$

Unlike ADE or FDE, valid time captures how long a trajectory remains practically trustworthy before its instantaneous positional error exceeds the threshold δ (i.e., drifting dangerously). A larger T_v demonstrates sustained predictive reliability, making it a paramount metric for real-time collision avoidance and autonomous navigation.

6 Experimental Results

6.1 UAV Trajectory Forecasting

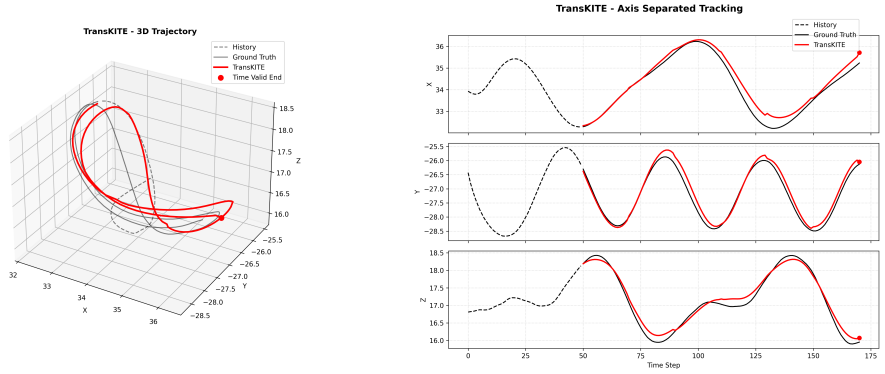
Table 2 demonstrates that TransKITE achieves the best performance across all metrics. By attaining the lowest ADE and RMSE, our method shows superior tracking accuracy. Combining the Transformer's global context with explicit kinematic constraints enables TransKITE to effectively capture both macroscopic trends and microscopic dynamics across the entire forecasting horizon.

Crucially, TransKITE achieves the highest valid time ($T_v = 107.52$ steps), remaining within the safety threshold significantly longer before diverging. This validates that the zero-input autonomous GRU decoder, regulated by Euler and integration constraints, successfully mitigates the exposure bias typical of standard autoregressive rollouts. TransKITE also establishes a new state-of-the-art in FDE. Long-horizon endpoint prediction is notoriously challenging due to compounding spatial uncertainty; TransKITE's superior FDE indicates it maintains physically bound predictive

Table 2: Comparison on the Synthetic UAV Flight Trajectories dataset. Results present the Mean ± Standard Deviation (5 seeds). Bold values indicate the best performance.

Model	ADE (m) ↓	RMSE (m) ↓	FDE (m) ↓	T_v (steps) ↑
DLinear	2.6638 ± 0.0363	2.9895 ± 0.0336	2.9292 ± 0.0369	29.32 ± 0.35
LSTM	2.9175 ± 0.0375	3.4063 ± 0.0374	4.1712 ± 0.0540	40.73 ± 0.49
Informer	2.5551 ± 0.0323	2.8902 ± 0.0323	3.3743 ± 0.0417	42.38 ± 0.51
PatchTST	2.7076 ± 0.0359	3.0657 ± 0.0349	3.4203 ± 0.0421	41.22 ± 0.50
NeuralODE	2.8771 ± 0.0367	3.0865 ± 0.0538	2.8407 ± 0.0368	48.99 ± 0.59
TrajCLIP	2.0151 ± 0.1016	2.3414 ± 0.1933	3.1047 ± 0.3825	67.08 ± 0.63
MS-TIP	1.8356 ± 0.1528	2.1020 ± 0.2467	2.5138 ± 0.3931	70.35 ± 1.32
MRGTraj	1.7996 ± 0.1061	2.0701 ± 0.1493	2.4262 ± 0.2120	69.06 ± 1.10
S-Implicit	1.6343 ± 0.1123	1.8818 ± 0.1193	2.4649 ± 0.1387	80.01 ± 1.30
SCA-LSTM	1.5816 ± 0.0710	1.8031 ± 0.0855	2.1004 ± 0.1010	77.47 ± 0.44
TransKITE (Ours)	$\mathbf{1.0178 \pm 0.0481}$	$\mathbf{1.1554 \pm 0.0517}$	$\mathbf{1.5213 \pm 0.0644}$	$\mathbf{107.52 \pm 4.36}$

capability until the final step without sacrificing short-term smoothness. Among baselines, trajectory-specific models like SCA-LSTM and S-Implicit perform relatively well, achieving sub-2.0m RMSEs by explicitly modeling 3D spatial dependencies. However, relying purely on data-driven objectives without strict mathematical constraints causes them to fall short of TransKITE over long horizons. Conversely, general-purpose Transformers (Informer, PatchTST) struggle with continuous UAV kinematics when applied naively to coordinate sequences. TransKITE overcomes this by restricting the Transformer to latent state encoding and delegating physical progression to the recurrent decoder. Finally, DLinear's high tracking errors reaffirm that purely linear mappings cannot model the highly nonlinear dynamics of 3D UAV flights.



(a) 3D UAV trajectory comparison.

(b) Per-axis temporal tracking comparison.

Fig. 2: Comparison between the ground truth and the predicted UAV trajectory.

Figure 2 provides a representative 3D UAV trajectory comparison. The solid red segment indicates where TransKITE's prediction remains strictly within the 2.0 m valid time threshold, ending at the red marker. Visually, TransKITE maintains an extended valid horizon that tightly adheres to the ground-truth loop. The per-axis decomposition in Figure 2b confirms consistent kinematic tracking across all coordinates before crossing the error threshold.

6.2 3DMotraj - UUV Trajectories

To evaluate generalizability, we extend our experiments to the 3DMoTraj dataset, which captures UUVs operating under distinct fluid dynamic resistance and inertia profiles. As summarized in Table 3, TransKITE achieves state-of-the-art performance with the lowest global mean ADE (0.50 m) and FDE (0.69 m). Notably, TransKITE reduces mean FDE by ∼32% compared to the strongest baselines (LSTM and SCA-LSTM, both at 1.02 m). This highlights the effectiveness of the zero-input GRU and absolute physics-informed constraints in mitigating compounding drift. TransKITE achieves the highest performance in five of the eight environments; while specialized models (SCA-LSTM, MRGTraj) gain marginal localized leads in specific scenes, TransKITE demonstrates remarkably superior global robustness.

Method	#1	#2	#3	#4	#5	#6	#7	#8	**Mean**
LSTM	0.58/0.96	0.40/0.63	0.51/0.80	1.08/1.92	0.36/0.55	**0.54**/0.58	0.71/1.23	0.84/1.48	0.63/1.02
Informer	0.63/0.97	0.45/0.64	0.61/0.98	0.83/1.45	0.35/0.54	0.76/0.97	0.93/1.40	0.80/1.49	0.67/1.05
PatchTST	0.38/0.79	0.42/0.74	0.46/0.83	1.02/2.02	0.40/0.71	0.75/1.01	0.89/1.90	0.89/1.73	0.65/1.22
DLinear	0.50/0.96	0.66/1.17	0.55/1.04	1.59/3.30	0.60/1.06	1.15/1.75	1.46/3.07	1.23/2.45	0.97/1.85
NeuralODE	0.41/0.60	0.39/0.63	0.48/0.78	1.03/1.92	0.37/0.53	**0.54/0.55**	1.99/3.64	0.82/1.46	0.75/1.26
TrajCLIP	0.56/0.94	0.40/0.69	0.37/0.70	1.32/2.81	0.33/0.58	0.79/1.27	1.04/2.02	0.85/1.69	0.71/1.34
MS-TIP	0.62/1.16	0.61/1.16	0.58/1.15	0.78/1.42	0.57/1.11	0.84/1.50	0.77/1.44	0.79/1.52	0.70/1.31
MRGTraj	0.63/1.29	**0.34/0.54**	0.33/0.55	0.94/1.89	0.42/0.83	0.99/1.73	1.06/2.44	0.84/1.63	0.69/1.36
S-Implicit	0.54/0.87	0.43/0.72	0.47/0.84	1.13/2.31	0.40/0.68	0.85/1.38	0.62/1.06	0.98/1.92	0.68/1.22
SCA-LSTM	**0.36/0.51**	0.37/0.60	0.48/0.86	0.81/1.69	0.28/0.44	0.69/1.10	1.12/1.95	0.55/0.99	0.58/1.02
TransKITE	0.66/0.98	0.46/0.60	**0.27/0.27**	**0.58/0.90**	**0.26/0.29**	0.68/0.91	**0.55/0.76**	**0.49/0.78**	**0.50/0.69**

Table 3: Comparison on the 3DMoTraj UUV dataset. Results are presented as ADE/FDE in meters. Bold values indicate the best performance, and underline values indicate the second-best performance.

The per-scene results further highlight the robustness of TransKITE across diverse UUV motion patterns. TransKITE achieves the best overall performance in five of the eight scenes, namely Scenes #3, #4, #5, #7, and #8. Trajectory-specific baselines such as SCA-LSTM and MRGTraj achieve competitive performance in some scenes, but their effectiveness varies considerably across the full benchmark. Sequence models such as Informer, PatchTST, and DLinear struggle to capture the highly coupled dynamics of UUV motion. Overall, these results show that TransKITE generalizes effectively beyond the UAV setting and achieves robust forecasting performance across diverse UUV dynamics.

6.3 Ablation Study

We conduct a comprehensive ablation study on the UAV dataset to examine the contribution of TransKITE's core design choices to forecasting accuracy and stability. Although conducted only in the UAV setting, this analysis provides a representative benchmark for understanding the behavior of the proposed framework in 3D trajectory forecasting. All experiments use identical training and evaluation settings.

We investigate three key aspects: input representation, velocity estimation, and physics-informed constraints. The "w/o velocity input" variant provides only 3D positions to the encoder, forcing it to implicitly infer dynamics. The "w/o Savitzky–Golay filtering" variant replaces polynomial smoothing with naive finite differences, testing vulnerability to high-frequency noise. Finally, we systematically disable the physics constraints: "Pure data" ($\beta = 0, \gamma = 0$), removing velocity-integration consistency ($\beta = 0$), and removing the Euler local-update ($\gamma = 0$).

Variant	**ADE (m)** ↓	$\Delta\%_{ADE}$	**RMSE (m)** ↓	**FDE (m)** ↓	T_v **(steps)** ↑	$\Delta\%_{T_v}$
TransKITE (Full Proposed)	$\mathbf{1.0178 \pm 0.0481}$	-	$\mathbf{1.1554 \pm 0.0517}$	$\mathbf{1.5213 \pm 0.0644}$	$\mathbf{107.52 \pm 4.36}$	-
w/o local-update constraint ($\gamma = 0$)	1.0346 ± 0.0489	+1.6%	1.1786 ± 0.0527	1.5237 ± 0.0645	105.70 ± 4.29	−1.7%
w/o consistency constraint ($\beta = 0$)	1.0572 ± 0.0502	+3.8%	1.2046 ± 0.0539	1.5992 ± 0.0677	105.33 ± 4.27	−2.0%
Pure data ($\beta = 0, \gamma = 0$)	1.0660 ± 0.0504	+4.7%	1.2120 ± 0.0542	1.6092 ± 0.0681	105.98 ± 4.30	−1.4%
w/o Savitzky–Golay filtering	1.1924 ± 0.0564	+17.1%	1.3624 ± 0.0610	1.8802 ± 0.0796	99.97 ± 4.05	−7.0%
w/o velocity input	2.1989 ± 0.1039	+116.0%	2.4296 ± 0.1087	2.7654 ± 0.1171	52.66 ± 2.14	−51.0%

Table 4: Comparison of different ablation variants of TransKITE. The $\Delta\%_{ADE}$ and $\Delta\%_{T_v}$ columns indicate the percentage degradation compared to the full model.

Table 4 shows all components act synergistically to extend trajectory stability. Removing velocity input causes catastrophic degradation: ADE, RMSE, and FDE more than double, and T_v plummets to 52.66 steps, indicating dual-kinematic inputs are indispensable for capturing complex dynamics. Replacing the Savitzky–Golay filter with finite differences drops T_v below 100 steps (99.97), highlighting that exposing the network to uncalibrated noise disrupts the latent dynamical state.

Crucially, ablating the physics-informed constraints confirms their necessity in mitigating autoregressive drift. The "Pure data" variant exhibits significantly higher spatial errors. Analyzing individual constraints, removing integration consistency degrades spatial accuracy (ADE: 1.0572) and valid time (T_v: 105.33) more than removing the step-wise local-update (ADE: 1.0346, T_v: 105.70). This indicates that while the Euler update regulates step-to-step coherence, the macroscopic integration constraint is slightly more dominant in maintaining long-term spatial bounds.

To further illustrate the impact of these components, Figure 3 visualizes the error accumulation trajectories over the prediction horizon. Specifically, the plotted values represent the mean instantaneous positional error (step-wise ADE) averaged across all samples in the test set at each prediction step k. As observed, while the ablated variants exhibit rapid spatial divergence as errors compound autoregressively, the full TransKITE model effectively bounds the step-wise drift, maintaining significantly lower mean errors across all future time steps.

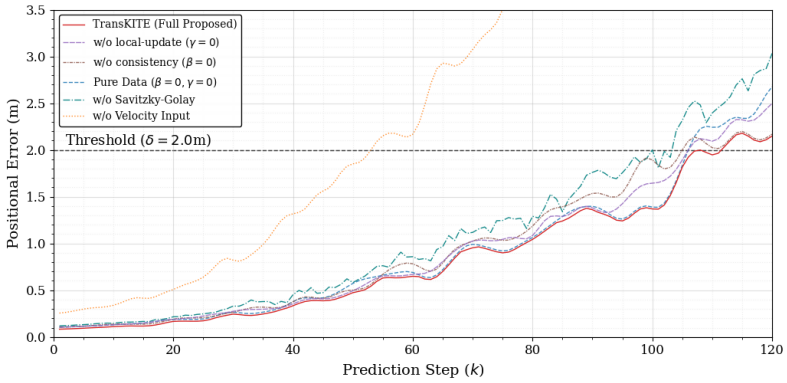


Fig. 3: Mean error accumulation trajectories over the prediction horizon. The instantaneous positional error at each step k is averaged across the entire test set.

6.4 Computational Complexity

For real-time autonomous systems, inference latency and memory footprint are as critical as predictive accuracy. To simulate onboard edge constraints, all benchmarks were conducted on a 16-core CPU server (32 GB RAM). For a fair comparison, all models share an identical hidden state dimension ($d_{model} = 64$).

As detailed in Table 5, TransKITE maintains a highly efficient memory footprint of 147,910 parameters. By restricting heavy Multi-Head Self-Attention to the history encoder and relying on a lightweight zero-input GRU for generation, TransKITE requires fewer parameters, trains faster, and executes faster than the full-attention Informer architecture (e.g., 87 ms vs. 104 ms inference in the UAV domain).

Table 5: Computational complexity and inference latency comparison.

Model	**Parameters**	**Training Time (min/epoch)**		**Inference Time (ms/seq)**	
		UAV	**UUV**	**UAV**	**UUV**
LSTM	75,512	0.89 ± 0.0198	0.54 ± 0.0121	41 ± 2	12 ± 1
Informer	168,262	2.32 ± 0.0516	1.46 ± 0.0327	104 ± 6	32 ± 2
Neural ODE	22,534	2.71 ± 0.0602	1.62 ± 0.0358	53 ± 3	17 ± 1
TransKITE	**147,910**	**1.85 ± 0.0329**	**1.07 ± 0.0343**	**87 ± 4**	**25 ± 1**

While simplified models (e.g., LSTM, Neural ODE) offer lower latencies, they lack the physical constraints to prevent severe trajectory drift. TransKITE strikes the optimal trade-off, delivering state-of-the-art accuracy, physical consistency, and extended valid time (T_v) while satisfying the real-time CPU processing constraints of modern autonomous controllers.

7 Discussion and Limitations

TransKITE's superior performance over state-of-the-art baselines highlights a critical paradigm shift: the necessity of embedding physical inductive biases into high-capacity neural architectures for trajectory forecasting. A persistent challenge in purely data-driven autoregressive modeling is exposure bias. While Transformers excel at global dependency modeling, they struggle to replicate continuous physical progression. TransKITE resolves this by delegating spatiotemporal context extraction to the Transformer encoder and physical progression to a zero-input autonomous GRU decoder. Isolated from noisy intermediate predictions, the decoder acts as a closed-loop dynamical system, naturally resisting step-wise error propagation.

Furthermore, our results demonstrate that merely feeding velocity into a neural network is insufficient for long-term stability; explicit physics-informed constraints are required to govern the output space. By enforcing a first-order Euler local-update and global velocity-integration consistency, TransKITE ensures that generated positions and velocities remain mathematically tethered. This hybrid approach - strictly bounding data-driven learning with kinematic laws - maintains physically plausible trajectories, evidenced by the substantial extension of valid prediction time (T_v).

Despite these improvements, we acknowledge certain limitations. First, while computationally efficient, the first-order Euler constraint may exhibit reduced accuracy during extreme acrobatic maneuvers characterized by sudden, massive accelerations. Second, generalizing across UAV and UUV domains currently necessitates domain-specific hyperparameter tuning due to distinct aerodynamic and hydrodynamic properties. Developing a domain-agnostic model with dynamic constraint adaptation and exploring higher-order integrators for extreme maneuverability remain compelling directions for future work.

8 Conclusion

In this paper, we introduced TransKITE, a novel Transformer-based Kinematics-Informed Trajectory Estimator for robust, long-horizon 3D UAV and UUV flight forecasting. Recognizing the drift vulnerabilities of standard autoregressive predictors, TransKITE fundamentally reshapes the pipeline by synergizing global attention mechanisms with autonomous dynamical decoding and physics-guided constraints.

Our approach augments noisy positional inputs with Savitzky–Golay-filtered velocity estimates, creating a refined 6D kinematic state. To prevent severe coordinate drift during long-term rollouts, the model is optimized via a hybrid objective enforcing discrete-time physical realism. Specifically, a first-order Euler local-update constraint ensures step-wise aerodynamic coherence, while a velocity-integration consistency constraint maintains macroscopic spatial bounds.

Extensive evaluations on 3D UAV and UUV trajectory datasets demonstrate that TransKITE establishes a new state-of-the-art. It achieves the lowest spatial errors while significantly increasing valid prediction time compared with a wide array of competitive baselines. Comprehensive ablations confirm the indispensability of each component: removing the kinematic input representation causes catastrophic degradation, while the global integration constraint proves slightly more critical than the

local-update in preventing irreversible error accumulation. Ultimately, our findings suggest that bridging advanced sequence modeling with fundamental physical inductive biases yields profound gains in predictive accuracy and operational stability. This positions TransKITE as a highly reliable framework for real-world autonomous systems, advancing capabilities in online trajectory planning and collision avoidance.

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Declarations

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- **Conflict of interest:** The authors have no relevant financial or non-financial interests to disclose.
- **Ethics approval:** Not applicable. This study does not involve human participants or animals.
- **Consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** The datasets analysed during the current study are publicly available. UAV: https://huggingface.co/datasets/riotu-lab/Synthetic-UAV-Flight-Trajectories. UUV: https://github.com/zhouhao94/3DMoTraj.
- **Code availability:** The source code, trained models, and configuration files required to reproduce the experimental results of the TransKITE framework are available at https://github.com/funzi-son/NeuroSciDyn.
- **Author contribution:** Son N. Tran conceived the research questions, established the overall research direction, and identified the data sources for evaluation. Minh Vu developed the initial methodology, produced the preliminary results, and prepared the first draft of the manuscript. G. Nam Luu introduced the transformer-based approach, conducted the theoretical analysis, and substantially revised and extended the paper. Binh Nguyen validates the methods and experimental results. All authors contributed to the analysis and interpretation of the methodology and experimental results and to the refinement and finalisation of the manuscript.

Appendix A Sensitivity Analysis of Window Size

To justify the selection of the sliding window size L for the Savitzky-Golay filter, we evaluated the model's tracking accuracy across different window sizes ($L \in \{5, 7, 11, 15\}$) on the UAV dataset. As shown in Table A1, a smaller window ($L = 5$) fails to sufficiently smooth the high-frequency measurement noise, while a larger window ($L = 15$) over-smooths the data, dampening the rapid kinematic transitions critical for accurate forecasting. The empirical choice of $L = 11$ provides the optimal trade-off, yielding the lowest displacement errors.

Window Size (L)	**ADE (m) ↓**	**RMSE (m) ↓**	**FDE (m) ↓**	T_v **(steps) ↑**
$L = 5$	1.0742 ± 0.0508	1.2210 ± 0.0547	1.6084 ± 0.0680	103.85 ± 4.22
$L = 7$	1.0417 ± 0.0493	1.1839 ± 0.0530	1.5586 ± 0.0659	106.10 ± 4.31
$\mathbf{L = 11}$	$\mathbf{1.0178} \pm 0.0481$	$\mathbf{1.1554} \pm 0.0517$	$\mathbf{1.5213} \pm 0.0644$	$\mathbf{107.52} \pm 4.36$
$L = 15$	1.0895 ± 0.0515	1.2378 ± 0.0555	1.6329 ± 0.0691	102.47 ± 4.16

Table A1: Sensitivity analysis of the sliding window size L on the UAV dataset.

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